

Name: _____ ID: _____ Section: _____

Question 1 [10 Points]

Write a Java program where you need to create a method called **"digitCalculator"**. The method takes two strings as parameters and calculates the sum of digits for both strings. Finally, the method returns the larger of the two sums of digits. If both of the sums are equal, return the sum of digits of the first string.

[Note: You are only allowed to use the built-in `length()`, `charAt()`, `codePointAt()` methods and any custom method you created to solve this problem.]

Sample Method Call 1: <code>int larger = digitCalculator("F4C5B3,4", "R0M2A2_3");</code> <code>System.out.println(larger);</code>	Sample Method Call 2: <code>int larger =digitCalculator("123", "56");</code> <code>System.out.println(larger);</code>
Sample Output 1: 16	Sample Output 2: 11
Explanation 2: In the first string, the digits are 4,5,3,4. The sum of digits is 16. In the second string, the digits are 0,2,2,3. The sum of digits is 7. So the larger sum was returned.	Explanation 2: In the first string, the digits are 1,2,3. The sum of digits is 6. In the second string, the digits are 5,6. The sum of digits is 11. So the larger sum was returned.

Question 2 [5 Points]

1	<code>public class Tracing5 {</code>
2	<code>public static void main(String[] args) {</code>
3	<code>int[] arr = {3, 4, 1, 2, 5};</code>
4	<code>recur(arr, 0, 0);</code>
5	<code>}</code>
6	<code>public static void recur(int[] b, int i, int s) {</code>
7	<code>if (i >= b.length) {</code>
8	<code>return;</code>
9	<code>}</code>
10	<code>int val = b[i];</code>
11	<code>b[i] = s;</code>
12	<code>recur(b, i + 1, s + val);</code>
13	<code>System.out.println(b[i]);</code>
14	<code>}</code>
15	<code>}</code>

Output